SMART AIR TRAFFIC CONTROL SYSTEM

C++ Source Code • Data Structures Project

```
#include <iostream>
#include <string>
using namespace std;

struct Aircraft
{
    int flightID;
    string airline;
    string destination;
    string operation;
    int runway;
    string status;

    Aircraft* next;
};

Aircraft* head = nullptr;

// Search for an aircraft
Aircraft* searchAircraftByID(int id)
{
    Aircraft* temp = head;

    while (temp != nullptr)
    {
        if (temp->flightID == id)
            return temp;

        temp = temp->next;
    }

    return nullptr;
}

// Add aircraft
void addAircraft()
{
    Aircraft* newNode = new Aircraft;

    cout << "Enter Flight ID: ";
    cin >> newNode->flightID;

    if (searchAircraftByID(newNode->flightID) != nullptr)
    {
        cout << "Flight ID already exists!\n";
        delete newNode;
        return;
    }

    cin.ignore(1000, '\n');

    cout << "Enter Airline Name: ";
    getline(cin, newNode->airline);

    cout << "Enter Destination: ";
    getline(cin, newNode->destination);

    cout << "Enter Operation (Landing/Departure): ";
    getline(cin, newNode->operation);

    cout << "Enter Runway Number: ";
    cin >> newNode->runway;

    cin.ignore(1000, '\n');

    cout << "Enter Status (Scheduled/Landing/Departing/Completed): ";
    getline(cin, newNode->status);

    newNode->next = nullptr;

    if (head == nullptr)
    {
        head = newNode;
    }
    else
```

```
    {
        Aircraft* temp = head;

        while (temp->next != nullptr)
            temp = temp->next;

        temp->next = newNode;
    }

    cout << "Aircraft added successfully!\n";
}

// Display all aircraft
void displayAircraft()
{
    if (head == nullptr)
    {
        cout << "No aircraft records available.\n";
        return;
    }

    Aircraft* temp = head;

    cout << "\n===== AIRCRAFT SCHEDULE =====\n";

    while (temp != nullptr)
    {
        cout << "\nFlight ID: " << temp->flightID;
        cout << "\nAirline: " << temp->airline;
        cout << "\nDestination: " << temp->destination;
        cout << "\nOperation: " << temp->operation;
        cout << "\nRunway: " << temp->runway;
        cout << "\nStatus: " << temp->status << "\n";

        temp = temp->next;
    }
}

// Search aircraft
void searchAircraft()
{
    int id;

    cout << "Enter Flight ID to search: ";
    cin >> id;

    Aircraft* temp = searchAircraftByID(id);

    if (temp == nullptr)
    {
        cout << "Aircraft not found.\n";
    }
    else
    {
        cout << "\nAircraft Found!\n";
        cout << "Flight ID: " << temp->flightID << "\n";
        cout << "Airline: " << temp->airline << "\n";
        cout << "Destination: " << temp->destination << "\n";
        cout << "Operation: " << temp->operation << "\n";
        cout << "Runway: " << temp->runway << "\n";
        cout << "Status: " << temp->status << "\n";
    }
}

// Update aircraft status
void updateStatus()
{
    int id;

    cout << "Enter Flight ID to update: ";
    cin >> id;

    Aircraft* temp = searchAircraftByID(id);

    if (temp == nullptr)
    {
        cout << "Aircraft not found.\n";
        return;
    }

    cin.ignore(1000, '\n');
```

```
    cout << "Enter new status: ";
    getline(cin, temp->status);

    cout << "Aircraft status updated successfully!\n";
}

// Delete aircraft
void deleteAircraft()
{
    int id;

    cout << "Enter Flight ID to delete: ";
    cin >> id;

    Aircraft* temp = head;
    Aircraft* previous = nullptr;

    while (temp != nullptr && temp->flightID != id)
    {
        previous = temp;
        temp = temp->next;
    }

    if (temp == nullptr)
    {
        cout << "Aircraft not found.\n";
        return;
    }

    if (previous == nullptr)
        head = temp->next;
    else
        previous->next = temp->next;

    delete temp;

    cout << "Aircraft record deleted successfully!\n";
}

// Free allocated memory
void clearRecords()
{
    while (head != nullptr)
    {
        Aircraft* temp = head;
        head = head->next;
        delete temp;
    }
}

int main()
{
    int choice;

    do
    {
        cout << "\n====================================\n";
        cout << " SMART AIR TRAFFIC CONTROL SYSTEM\n";
        cout << "====================================\n";
        cout << "1. Add Aircraft\n";
        cout << "2. Display Aircraft Schedule\n";
        cout << "3. Search Aircraft\n";
        cout << "4. Update Aircraft Status\n";
        cout << "5. Delete Aircraft\n";
        cout << "6. Exit\n";
        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice)
        {
            case 1:
                addAircraft();
                break;

            case 2:
                displayAircraft();
                break;

            case 3:
                searchAircraft();
                break;
```

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            case 4:
                updateStatus();
                break;

            case 5:
                deleteAircraft();
                break;

            case 6:
                clearRecords();
                cout << "System closed successfully.\n";
                break;

            default:
                cout << "Invalid choice! Please try again.\n";
        }

    } while (choice != 6);

    return 0;
}
```